

Eisenstein Series Modulo Prime Powers

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International Seminar on Automorphic Forms

May 5, 2026

Eisenstein Series for $SL_2(\mathbb{Z})$

Recall, for $k \in \mathbb{Z}_{\geq 2}$ even:

Definition

$$G_k := \frac{-B_k}{2k} E_k := \frac{-B_k}{2k} + \sum_{n \geq 1} \sigma_{k-1}(n) q^n.$$

1. B_k from $\frac{x}{e^x - 1} =: \sum_{k \geq 0} B_k \frac{x^k}{k!}$ the k th Bernoulli number
2. $\sigma_j(n) := \sum_{d|n} d^j$
3. Define $E_0 := 1$.

Basic Theory of Eisenstein Series

1. If $k \geq 4$ then E_k is a holomorphic modular form of weight k for $\mathrm{SL}_2(\mathbb{Z})$
2. If $k = 2$ then E_2 is *quasimodular*:

$$E_2\left(\frac{a\tau + b}{c\tau + d}\right) = (c\tau + d)^2 E_2(\tau) - \frac{6i}{\pi} c(c\tau + d)$$

for $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z})$

Classical Congruences of Eisenstein Series

$p \geq 5$ a prime

p -integrality

G_k is p -integral^a $\iff (p-1) \nmid k$

$$^a G_k \in M_k \cap \mathbb{Z}_{(p)}[[q]]$$

- ▶ p -integrality depends on the constant coefficient of $\frac{-B_k}{2k} + \sum_{n \geq 1} \sigma_{k-1}(n)q^n$
- ▶ Clausen von-Stadt: $B_k \equiv - \sum_{\substack{q \text{ prime} \\ (q-1) \mid k}} \frac{1}{q} \pmod{1}$
- ▶ $v_p\left(\frac{B_k}{k}\right) \geq 0$ for $(p-1) \nmid k$
- ▶ $v_p\left(\frac{B_k}{k}\right) = -v_p(k) - 1$ else

Classical Congruences of Eisenstein Series

$p \geq 5$ a prime

- ▶ G_k is p -integral $\iff (p-1) \nmid k$

Useful Fact

$$G_k \equiv G_{k'} \pmod{p} \text{ if } k \equiv k' \not\equiv 0 \pmod{p-1}$$

- ▶ $E_k \equiv 1 \pmod{p}$ if $(p-1) \mid k$
- ▶ If

$$\partial f = 12q \frac{d}{dq} f + 24k G_2 f$$

is the Ramanujan-Serre derivative¹ $\partial : M_k \rightarrow M_{k+2}$ then

$$\partial E_{p-1} \equiv E_{2+(p-1)} \equiv E_2 \pmod{p}$$

¹Correcting a typo in a previous version

What about higher power congruences?

Classical Prime Power Congruences

Generalizing the useful fact

Let $(p-1) \nmid k_0$ and $k_0 > m$, then

$$G_{k_0} \equiv G_{p^{m-1}(p-1)+k_0} \pmod{p^m}.$$

- ▶ Kummer Congruences $\implies$ if $(p-1) \nmid k$ and $k \equiv k' \pmod{p^m(p-1)}$ then

$$(1 - p^{k-1}) \frac{B_k}{k} \equiv (1 - p^{k'-1}) \frac{B_{k'}}{k'} \pmod{p^m}.$$

Deals with the constant term.

- ▶ Fermat-Euler theorem: $\sigma_{k_0-1}(n) \equiv \sigma_{k-1}(n) \pmod{p^m}$ resolves the coefficient of q^n .

A natural question

Question: How low a weight is a modular form $f \in M_k$ determined modulo p^m ?

Definition

The weight filtration ω_{p^m} of $f \in M_k \cap \mathbb{Z}_{(p)}[[q]]$ is

$$\omega_{p^m}(f) := \inf\{k' : f \equiv g \pmod{p^m} \text{ for some } g \in M_{k'}\}.$$

Our generalized fact $\implies \omega_{p^m}(G_k) \leq m + p^{m-1}(p-1),$
 $(p-1) \nmid k.$

But can we do better?

Prior Work

Recall $E_{p-1} \equiv 1 \pmod{p}$.

Theorem 1.1 (Ahlgren Hanson Raum Richter, 2024)

Let $k \equiv k_0 \pmod{p-1}$, $2 \leq k_0 \leq p-3$. There is an $f_{(p-1)+k_0} \in M_{(p-1)+k_0} \cap \mathbb{Z}_{(p)}[[q]]$ such that

$$G_k \equiv f_{(p-1)+k_0} E_{p-1}^n \pmod{p^2}.$$

$$n = \frac{k-k_0}{p-1} - 1.$$

Note: $(p-1) + k_0 < 2 + p(p-1)$.

Prior Work

Recall $E_{p-1} \equiv 1 \pmod{p}$.

E_k as Pure Powers (AHRR, 2024)

For any $p \geq 5$ prime we have

$$E_{rp^{k-1}(p-1)} \equiv E_{p-1}^{rp^{k-1}} \pmod{p^{k+1}}.$$

AHRR Proof Method

Popa's Identity (2011)

For $p \geq 5$ a prime and $k \in \mathbb{Z}_{>0}$, $1 \leq m \leq 2k - 2$ let

$$\tilde{m} := 2k - 2 - m$$

$$F_m := \sum_{\substack{j=3 \\ j \text{ odd}}}^m \binom{m}{j} G_{j+1} G_{\tilde{j}+1}.$$

Then is an explicit rational $c(k, m)$ with

$$c(k, m) G_{2k} = \frac{1}{24} \partial G_{2k-2} + F_m - G_{j+1} G_{\tilde{j}+1} + F_{\tilde{m}}.$$

AHRR Proof Method

$$G_k \equiv f_{(p-1)+k_0} E_{p-1}^n \pmod{p^2}$$

Key idea: Theorem is true for $G_{k_0+(p-1)}$, so we can proceed by induction on n in

$$G_{k_0+(n+1)(p-1)}$$

together with careful p -adic analysis. At each step we must pick correct parameters for m in Popa's identity.

Refining the Weight Filtration

Definition

The factor filtration $\tilde{\omega}_{p^m}$ of $f \in M_k \cap \mathbb{Z}_{(p)}[[q]]$ is

$$\tilde{\omega}_{p^m}(f) := \inf\{k' : f \equiv gE_{p-1}^n \pmod{p^m} \\ \text{for some } g \in M_{k'} \text{ and some } n \geq 0\}.$$

Examples

1. (Trivial) $f \in M_k$ then $\tilde{\omega}_p(f) = \tilde{\omega}_p(fE_{p-1}^n) \leq k$ for all n, k
2. (AHRR) $G_{284} = G_{32+7 \times 36} \equiv f_{68}E_{36}^6$ for explicit $f_{68} \in M_{68}$, so

$$\tilde{\omega}_{37^2}(G_{284}) \leq 68$$

Refining the Weight Filtration

Corollary (Ahlgren Hanson Raum Richter 2024)

$$\tilde{\omega}_{p^2}(G_k) \leq (p-1) + k_0,$$

$$k \equiv k_0 \pmod{p-1}, 2 \leq k_0 \leq p-3.$$

Example (AHRR)

$$G_{284} = G_{32+7 \times 36} \equiv f_{68} E_{36}^6:$$

$$\tilde{\omega}_{37^2}(G_{284}) \leq 68$$

$$k_0 = 32 \text{ and } 68 = 32 + 36$$

Our Result

Main Theorem I (Ahlgren, Castillio, W. 2025)

Define

$$H(m, \alpha, r) := (-1)^{m+r+1} \binom{\alpha - r - 1}{m - r - 1} \binom{\alpha}{r}$$

for $0 \leq r \leq m - 1$. Let $k^* > m$ be such that $(p - 1) \nmid k^*$.
Then

$$G_{\alpha(p-1)+k^*} \equiv \sum_{r=0}^{m-1} H(m, \alpha, r) G_{r(p-1)+k^*} E_{p-1}^{\alpha-r} \pmod{p^{\textcolor{brown}{m}}}.$$

Our Result

$$G_{\alpha(p-1)+k^*} \equiv \sum_{r=0}^{m-1} H(m, \alpha, r) G_{r(p-1)+k^*} E_{p-1}^{\alpha-r} \pmod{p^m}.$$

1. All terms have the same weight
2. $H(m, \alpha, r) = \delta_{r, \alpha}$ for $0 \leq \alpha \leq m-1$
3. $m = 1$ gives the classical congruence $G_k \equiv G_{k'} \pmod{p}$ if $k \equiv k' \not\equiv 0 \pmod{p-1}$
4. $m = 2$ and $k_0 \geq 4$ gives AHRR Theorem 1.1; does not work for $k_0 = 2$

Our Result

Main Theorem II (Ahlgren, Castillio, W. 2025)

Let $1 \leq m \leq p-1$, $\alpha \geq 1$. Then

$$E_{\alpha(p-1)} \equiv \sum_{r=0}^{m-1} H(m, \alpha, r) E_{r(p-1)} E_{p-1}^{\alpha-r} \pmod{p^m}.$$

Possible Generalization?

If $k^* > m$ is a multiple of $p-1$ then we should have

$$E_{\alpha(p-1)+k^*} \equiv \sum_{r=0}^{m-1} H(m, \alpha, r) E_{r(p-1)+k^*} E_{p-1}^{\alpha-r} \pmod{p^m}.$$

A Consequence of Our Result

Corollary (ACW, 2025)

Let $k_0(m) :=$ the smallest integer greater than m and congruent to $k \pmod{p-1}$. Then

$$\tilde{\omega}_{p^m}(G_k) \leq (m-1)(p-1) + k_0(m)$$

$$k > m; k_0(m) \leq m + p - 1$$

$$\implies \tilde{\omega}_{p^m}(G_k) \leq mp$$

An Example

Let

▶ $p = 7, m = 8$

▶ $k = 337(p - 1) + 4 = 2026$

Then $k_0(m) = 10$ and

$$(m - 1)(p - 1) + k_0(m) = 52.$$

There is an (explicit) $f_{52} \in M_{52}$ with

$$G_k = G_{2026} \equiv f_{52} E_6^{329} \pmod{7^8}$$

and this result is *sharp*; there is no $g \in M_{46}$ with

$$f_{52} \equiv g E_6 \pmod{7^8}.$$

An Example

$$\begin{aligned} f_{52} = 289118E_4^{13} + 3330770E_4^{10}\Delta + 1615995E_4^7\Delta^2 \\ + 4467661E_4^4\Delta^3 + 1172952E_4\Delta^4 \in M_{52}. \end{aligned}$$

Another Example

Let

- ▶ $p = 17, m = 6$
- ▶ $k = 81(p - 1) = 1296$

There is an (explicit) $f_{80} \in M_{80}$ with

$$E_k = E_{1296} \equiv f_{80} E_{16}^{76} \pmod{17^6}$$

and this result is *sharp* (there is no smaller weight modular form satisfying the factor filtration condition.)

An Example

$$f_{80} = E_4^{20} + 17835578E_4^{17}\Delta + 1427399E_4^{14}\Delta^2 + 23585491E_4^{11}\Delta^3 \\ + 19629555E_4^8\Delta^4 + 23614096E_4^5\Delta^5 + 44217E_4^2\Delta^6 \in M_{80}.$$

Ideas in the Proof

Zhi-Hong Sun (1997, 2000):

1. $h : \mathbb{Z}_{\geq 0} \rightarrow \mathbb{Z}_{(p)}$ is p -regular if

$$\sum_{k=0}^m \binom{m}{k} (-1)^k h(k) \equiv 0 \pmod{p^m}$$

for all m .

2. The product of p -regular functions is p -regular
3. If h is p -regular then

$$h(\alpha) \equiv \sum_{r=0}^{m-1} H(m, \alpha, r) h(r) \pmod{p^m}$$

for all α, m .

Ideas in the Proof

4. Prove the claim

$$G_{\alpha(p-1)+k^*} \equiv \sum_{r=0}^{m-1} H(m, \alpha, r) G_{r(p-1)+k^*} \pmod{p^m}$$

using p -regularity of $k \mapsto d^{k(p-1)}$ in σ divisor sums.

5. Write $E_{p-1} = 1 + pE$ and expand so the right hand side of the Main Theorem I becomes

$$\begin{aligned} & \sum_{r=0}^{m-1} H(m, \alpha, r) G_{r(p-1)+k^*} E_{p-1}^{\alpha-r} \\ & \equiv \sum_{j=0}^{m-1} p^j E^j \sum_{r=0}^{m-1} \binom{\alpha-r}{j} H(m, \alpha, r) G_{r(p-1)+k^*} \pmod{p^m} \end{aligned}$$

Our claim shows that the $j = 0$ term is $\equiv G_{\alpha(p-1)+k^*}$.

Ideas in the Proof

$$\begin{aligned} & \sum_{r=0}^{m-1} H(m, \alpha, r) G_{r(p-1)+k^*} E_{p-1}^{\alpha-r} \\ & \equiv \sum_{j=0}^{m-1} p^j E^j \sum_{r=0}^{m-1} \binom{\alpha-r}{j} H(m, \alpha, r) G_{r(p-1)+k^*} \pmod{p^m} \end{aligned}$$

6. Use p -regularity to write

$$\begin{aligned} & \sum_{r=0}^{m-1} \binom{\alpha-r}{j} H(m, \alpha, r) G_{r(p-1)+k^*} \\ & \equiv \sum_{s=0}^{m-j-1} G_{s(p-1)+k^*} \sum_{r=0}^{m-1} \binom{\alpha-r}{j} H(m, \alpha, r) H(m-j, r, s) \\ & \pmod{p^{m-j}} \end{aligned}$$

Ideas in the Proof

7. It suffices to prove

$$\sum_{r=s}^{m-1} \binom{\alpha - r}{j} H(m, \alpha, r) H(m - j, r, s) = 0$$

for $j \geq 1$.

Ideas in the Proof

Let

$$S(m) := \sum_{r=s}^{m-1} \binom{\alpha - r}{j} H(m, \alpha, r) H(m - j, r, s).$$

We get

$$(\alpha - m)S(m) + (m - s)S(m + 1) = 0,$$

which means we only need to prove $S(s + j + 1) = 0$, which proves the main theorem.

Ideas in the Proof

$$S(m) := \sum_{r=s}^{m-1} \binom{\alpha - r}{j} H(m, \alpha, r) H(m - j, r, s)$$

want $S(s + j + 1) = 0$

8.

$$S(s + j + 1) = \sum_{r=s}^j (-1)^{r+j+s} \binom{\alpha - r}{j} \binom{\alpha - 1 - r}{s + j - r} \binom{\alpha}{r} \binom{r}{s}$$

9.

$$\alpha > s + j \implies S(s + j + 1) = 0$$

10. Else

$$S(s + j + 1) \doteq {}_2F_1(-j, j - \beta; 1 - \beta; 1) = 0$$

Ideas in the Proof: $(p - 1) \mid k$

$$E_{\alpha(p-1)} \equiv \sum_{r=0}^{m-1} H(m, \alpha, r) E_{r(p-1)} E_{p-1}^{\alpha-r} \pmod{p^m}.$$

When $(p - 1) \mid k$ we use similar ideas, but we also need a congruence of Bernoulli numbers.

Proposition (ACW, 2025)

Let $\alpha \geq 1$ and $1 \leq m \leq p - 1$. then for any positive $d \in \mathbb{Z}$ with $p \nmid d$

$$d^{\alpha(p-1)} \frac{\alpha}{B_{\alpha(p-1)}} \equiv \sum_{r=1}^{m-1} H(m, \alpha, r) d^{r(p-1)} \frac{r}{B_{r(p-1)}} \pmod{p^m}.$$

Ideas in the Proof: $(p - 1) \mid k$

$$d^{\alpha(p-1)} \frac{\alpha}{B_{\alpha(p-1)}} \equiv \sum_{r=1}^{m-1} H(m, \alpha, r) d^{r(p-1)} \frac{r}{B_{r(p-1)}} \pmod{p^m}.$$

Key Idea:

1. Let $f(k) := (p - p^{k(p-1)})B_{k(p-1)}$
2. Cook up: g such that $f + g$ is p -regular and
- 3.

$$d^{\alpha(p-1)} \frac{p\alpha}{f(\alpha) + g(\alpha)} \equiv \sum_{r=1}^{m-1} H(m, \alpha, r) d^{r(p-1)} \frac{pr}{f(r) + g(r)}$$

4.

$$\frac{pr}{f(r) + g(r)} \equiv \frac{r}{B_{r(p-1)}} \pmod{p^m}$$

Refinements Modulo p^3

Proposition (ACW 2025)

Let $p \geq 5$ be prime and write $k \geq 4$ as $k = \alpha(p - 1) + k_0$ with $2 \leq k_0 \leq p - 3$.

1. If $k_0 \geq 4$ then

$$\tilde{\omega}_{p^3}(G_k) \leq \begin{cases} (p - 1) + k_0, & \text{if } \alpha \equiv 0, 1 \pmod{p}; \\ 2(p - 1) + k_0, & \text{otherwise.} \end{cases}$$

2. If $k_0 = 2$ then

$$\tilde{\omega}_{p^3}(G_k) \leq \begin{cases} (p - 1) + 2, & \text{if } \alpha \equiv 1 \pmod{p}; \\ 2(p - 1) + 2, & \text{if } \alpha \equiv 2 \pmod{p}; \\ 3(p - 1) + 2, & \text{otherwise.} \end{cases}$$

Proof Sketch (mod p^3)

$k_0 \geq 4$:

1. General: $\tilde{\omega}_{p^3}(G_k) \leq 2(p-1) + k_0$
2. Non-Archimedean: $\tilde{\omega}_{p^m}(f+g) \leq \max\{\tilde{\omega}_{p^m}(f), \tilde{\omega}_{p^m}(g)\}$
3. $\tilde{\omega}_{p^m}(pf) = \tilde{\omega}_{p^{m-1}}(f)$

Key idea: Write

$$G_{\alpha(p-1)+k_0} \equiv \binom{\alpha-1}{2} G_{k_0} E_{p-1}^\alpha - \alpha(\alpha-2) G_{(p-1)+k_0} E_{p-1}^{\alpha-1} \\ + \binom{\alpha}{2} G_{2(p-1)+k_0} E_{p-1}^{\alpha-2} \pmod{p^3}.$$

When $\alpha \equiv 0, 1 \pmod{p}$ then $\binom{\alpha}{2} \equiv 0 \pmod{p}$.

Proof Sketch (mod p^3)

Then

$$\begin{aligned}\tilde{\omega}_{p^3}(G_{\alpha(p-1)+k_0}) &\leq \max \{ (p-1) + k_0, \tilde{\omega}_{p^2}(G_{2(p-1)+k_0}) \} \\ &= (p-1) + k_0.\end{aligned}$$

Refinements Modulo p^4

Proposition (ACW 2025)

Let $p \geq 5$ be prime and write $k \geq 4$ as $k = \alpha(p - 1) + k_0$ with $2 \leq k_0 \leq p - 3$.

1. If $k_0 \geq 6$ then

$$\tilde{\omega}_{p^4}(G_k) \leq \begin{cases} (p - 1) + k_0, & \text{if } \alpha \equiv 0, 1 \pmod{p^2}; \\ 2(p - 1) + k_0, & \text{if } \alpha \equiv 0, 1, 2 \pmod{p}; \\ 3(p - 1) + k_0, & \text{otherwise.} \end{cases}$$

Refinements Modulo p^4

Proposition (ACW 2025)

Let $p \geq 5$ be prime and write $k \geq 4$ as $k = \alpha(p - 1) + k_0$ with $2 \leq k_0 \leq p - 3$.

2. If $k_0 = 4$ then

$$\tilde{\omega}_{p^4}(G_k) \leq \begin{cases} (p - 1) + 4, & \text{if } \alpha \equiv 1 \pmod{p^2}; \\ 2(p - 1) + 4, & \text{if } \alpha \equiv 1, 2 \pmod{p}; \\ 3(p - 1) + 4, & \text{if } \alpha \equiv 3 \pmod{p}; \\ 4(p - 1) + 4, & \text{otherwise.} \end{cases}$$

Refinements Modulo p^4

Proposition (ACW 2025)

Let $p \geq 5$ be prime and write $k \geq 4$ as $k = \alpha(p - 1) + k_0$ with $2 \leq k_0 \leq p - 3$.

3. If $k_0 = 2$ then

$$\tilde{\omega}_{p^4}(G_k) \leq \begin{cases} (p - 1) + 2, & \text{if } \alpha \equiv 1 \pmod{p^2}; \\ 2(p - 1) + 2, & \text{if } \alpha \equiv 2 \pmod{p^2}; \\ 3(p - 1) + 2, & \text{if } \alpha \equiv 1, 2, 3 \pmod{p}; \\ 4(p - 1) + 2, & \text{otherwise.} \end{cases}$$

Related Results and Possible Future Work

Our result when $(p-1) \mid k$ is limited to p^m , $1 \leq m \leq p-1$.

We expect: for $k^* > m$ a multiple of $(p-1) \mid k^*$ we have

$$E_{\alpha(p-1)+k^*} \equiv \sum_{r=0}^{m-1} H(m, \alpha, r) E_{r(p-1)+k^*} E_{p-1}^{\alpha-r} \pmod{p^m}.$$

Follows from this congruence of Bernoulli numbers:

Requires:

Let $\alpha \geq 1$. Then for any positive $d \in \mathbb{Z}$ with $p \nmid d$ we have

$$\frac{\alpha(p-1) + k^*}{B_{\alpha(p-1)+k^*}} \equiv \sum_{r=0}^{m-1} H(m, \alpha, r) \frac{r(p-1) + k^*}{B_{r(p-1)+k^*}} \pmod{p^m}.$$

Related Results and Possible Future Work

Theorem 1.2 AHRR (2024)

Let $p \geq 5$ be prime. Then

$$E_2 \equiv E_{p-1}^{p-1} \partial E_{p-1} + p E_{(p-1)+2}^p \pmod{p^2}.$$

Thank you!

Related Results and Possible Future Work

Why care?

Mazur's Eisenstein Ideal (Torsion Theorem)

Determines torsion part of an elliptic curve $E(\mathbb{Q})$.

Modular Forms modulo p (Swinnerton-Dyer and Katz)

$E_{p-1} - 1$ generates the ideal I making $\mathbb{F}_p[X, Y]/I \cong \sum_k M_k \pmod{p}$.

Serre's Conjecture (Kisin; Khare and Wintenberger)

An odd, irreducible, 2-dimensional Galois representation over a finite field arises from a modular form.